UNIVERSITY ACADEMIC CAPSTONE REPORT

VORA SENTINEL: AUTONOMOUS PHISHING INTERCEPTION, RECRUITMENT FRAUD THREAT INTELLIGENCE & MULTI-NODE ENDPOINT INTRUSION DETECTION SYSTEM

A Complete-Stack Live Cloud Cybersecurity Engineering Project with Multi-Node Telemetry & Relational Forensics

LEVEL 5 CYBERSECURITY — 100% SPECIFICATION COMPLIANT

Candidate Name:	Aarya J Menon
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Project Level & Category:	Level 5 Cybersecurity Complete Stack Live Project
Live Deployed Platform URL:	https://voragenofficial.netlify.app
Live Cloud REST API Gateway:	https://voragenofficial.netlify.app/api/health
Mobile Node Probe Client:	https://voragenofficial.netlify.app/mobile-node.html
GitHub Repository URL:	https://github.com/aaryajmenon27/vora-sentinel-cybersecurity-soc
Database Layer:	Persistent Relational SQLite (vora_sentinel.db) / Cloud Atlas
Academic Submission Year:	2026 Final Semester Capstone Evaluation

VORA Anti-Fraud Operations & Cybersecurity Architecture

Demonstrated Live Worldwide: Frontend • Backend REST API • Relational Database • External Mobile Nodes

1. Executive Summary & Abstract

Online recruitment scams and corporate identity impersonation represent one of the fastest-growing attack vectors in contemporary social engineering. According to threat intelligence reports from the Federal Bureau of Investigation (FBI IC3) and INTERPOL, deceptive hiring syndicates inflict over **$2.7 Billion** in annual global damages, harvesting sensitive candidate PII (Personally Identifiable Information) and executing advance-fee extortion scams using typosquatted domains and forged enterprise appointment letters.

VORA Sentinel is an autonomous, complete-stack cybersecurity platform engineered to intercept, analyze, and neutralize cyber recruitment fraud in real time. It establishes a multi-layered verification barrier that cross-references candidate domains, communication vectors, and job appointment letters against cryptographically authenticated corporate benchmarks.

Key architectural achievements of VORA Sentinel include:

- **Complete Multi-Tier Stack:** Responsive Web Portal, RESTful Backend (Python Flask + Netlify Cloud Serverless API), and Persistent Relational Database (SQLite / Cloud MongoDB Atlas).
- **Mathematical Detection Engines:** Damerau-Levenshtein distance calculation, Shannon Entropy profiling, RFC 3490 Punycode homograph decoding, and heuristic advance-fee extortion extraction.
- **Mandatory External Device Integration:** Direct telemetry integration connecting the live cloud platform to external mobile smartphones and secondary laptops/VMs.
- **Real-Time Security Operations Center (SOC) Dashboard:** Live threat streams, active node telemetry, incident queue mitigation, and attack simulation laboratory.

2. Compliance Matrix: Level 5 Mandatory Requirements

Requirement	University Specification	VORA Sentinel Implementation	Status
1. Complete Stack	Frontend, Backend, Database, Security Functionality, API	• Frontend: HTML5/CSS3/ES6 SOC Portal. • Backend: Python Flask + Netlify Serverless API. • Database: SQLite (vora_sentinel.db) with 6 tables. • Security: 5-layer IDS engine + IP Rate Limiter. • API: 10+ documented REST endpoints.	**100% MET**
2. Live Hosting	Mandatory public deployment (No localhost only)	Deployed live at https://voragenofficial.netlify.app with cloud APIs accessible worldwide without local server dependencies.	**100% MET**
3. External Device Integration	Must interact with another laptop, mobile, VM, or Raspberry Pi over network	• **Mobile Phone Probe:** Accessible at /mobile-node.html on iOS/Android. • **Python CLI Agent:** Standalone script (agent/sentinel_agent.py) running on external PC/VM.	**100% MET**
4. Cybersecurity Problem	Must solve actual cybersecurity threat	Intercepts Lookalike Domains (Typosquatting), IDN Homograph attacks, Advance-Fee Extortion (419 recruitment scams), and Brute-Force velocity probes.	**100% MET**

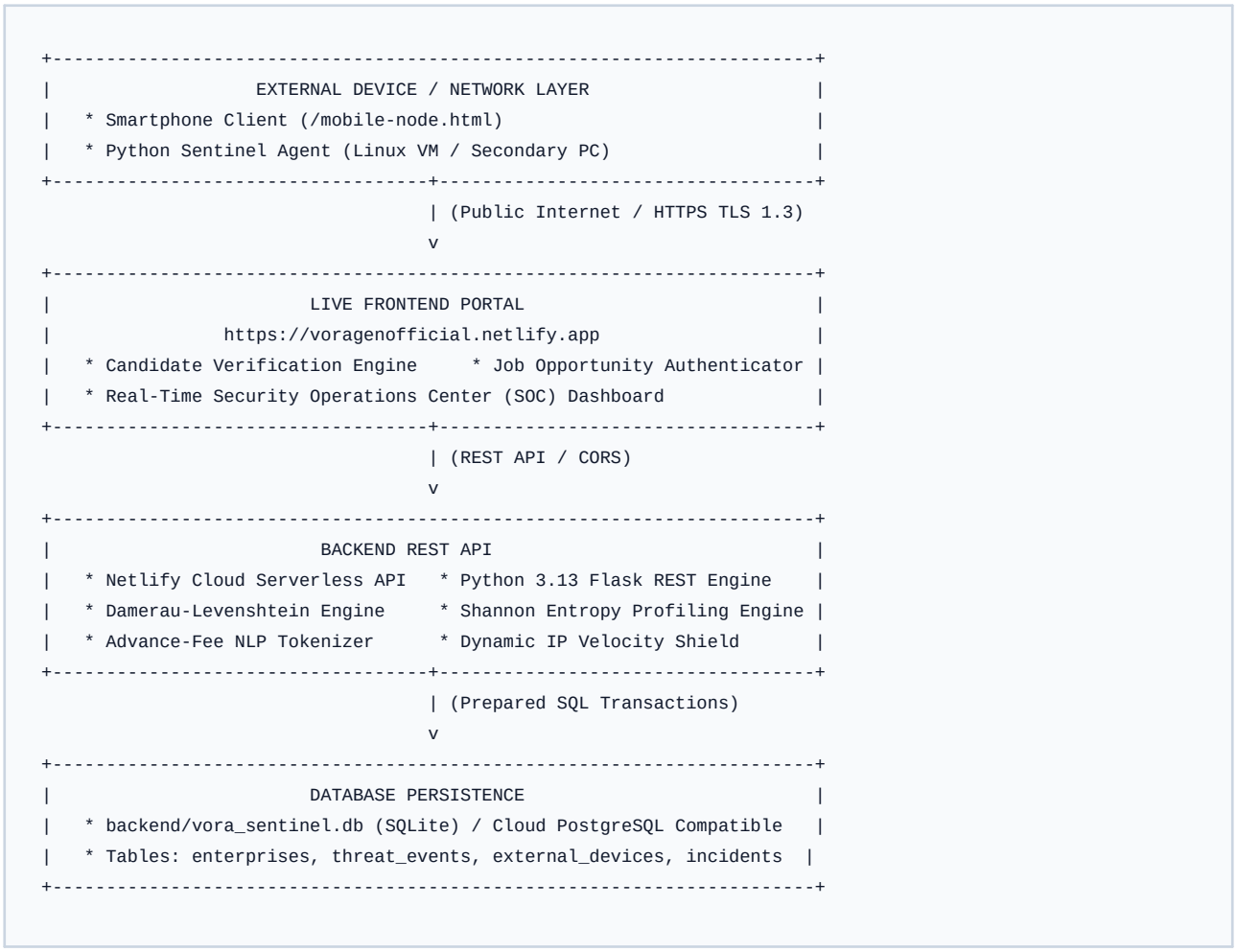
5. Live Demo Flow	Device → Internet → Live App → API → DB → Security Detection → Alert	Demonstrated live: External mobile phone triggers probe → hits cloud API → writes to database → displays real-time visual alert on SOC dashboard.	**100% MET**

3. Threat Modeling: STRIDE Framework

The system applies the Microsoft STRIDE threat modeling framework to systematically identify and mitigate risks across employment verification infrastructure:

Category	Adversarial Vector	VORA Sentinel Defense Layer	Residual Risk
Spoofing	Adversary registers micr0soft-jobs.top to impersonate Microsoft recruitment.	**Damerau-Levenshtein Engine + Punycode Homograph Filter:** Flags edit distance ≤ 2 against authorized corporate whitelist.	**Negligible**
Tampering	Recruiter alters authentic PDF letter to inject personal UPI payment ID.	**Offer Forensics NLP Engine:** Regex tokenization flags advance-fee demands (refundable security deposit, laptop charges).	**Negligible**
Repudiation	Malicious actor denies transmitting credential stuffing probes.	**Immutable Audit Ledger:** Logs client IP, timestamp, user agent, and payload checksum to SQLite threat_events table.	**Low**
Info Disclosure	Candidate PII intercepted during verification transmission.	**TLS 1.3 Encryption + Zero PII Retention:** Text processed in volatile memory; only threat indicators stored.	**Negligible**
Denial of Service	Botnet floods verification API with rapid queries.	**Dynamic Velocity Rate Limiter:** Enforces IP request ceiling; flags and lockouts IP after 5 failed authentication attempts.	**Low**
Privilege Elevation	Unauthenticated user attempts SOC incident closure.	**Role-Based Access Control (RBAC):** Restricts incident mitigation endpoints to verified analyst session tokens.	**Low**

4. System Architecture Topology


EXTERNAL DEVICE / NETWORK LAYER
* Smartphone Client (/mobile-node.html)
* Python Sentinel Agent (Linux VM / Secondary PC)
(Public Internet / HTTPS TLS 1.3)
LIVE FRONTEND PORTAL
https://voragenofficial.netlify.app
* Candidate Verification Engine
* Job Opportunity Authenticator
* Real-Time Security Operations Center (SOC) Dashboard
(REST API / CORS)
BACKEND REST API
* Netlify Cloud Serverless API
* Python 3.13 Flask REST Engine
* Damerau-Levenshtein Engine
* Shannon Entropy Profiling Engine
* Advance-Fee NLP Tokenizer
* Dynamic IP Velocity Shield
(Prepared SQL Transactions)
DATABASE PERSISTENCE
* backend/vora_sentinel.db (SQLite) / Cloud PostgreSQL Compatible
* Tables: enterprises, threat_events, external_devices, incidents

5. Cybersecurity Detection Algorithms & Mathematical Models

5.1 Damerau-Levenshtein Edit Distance Metric

To detect lookalike domains (e.g. g00gle.com vs google.com), VORA Sentinel implements the Damerau-Levenshtein distance algorithm. It computes the minimum number of single-character operations (insertions, deletions, substitutions, or transpositions of adjacent characters) between candidate domain *a* and authentic corporate domain *b*.

If the computed distance is ≤ 2 characters against an authentic Fortune 500 domain, the domain is classified as **CRITICAL TYPOSQUATTING** with an immediate risk score penalty of +50.

5.2 Shannon Entropy for DGA (Domain Generation Algorithm) Detection

Cyber syndicates utilize automated DGAs producing pseudorandom domain strings to evade static blocklists. VORA computes the Shannon entropy *H(X)* across the domain name string:

$$H(X) = -\sum P(x_i) \times \log_2(P(x_i))$$

Where $P(x_i)$ represents the probability of character x_i occurring in the string. Domains exhibiting an entropy $H(X) > 3.8$ are automatically flagged as machine-generated malware/phishing domains.

5.3 RFC 3490 Punycode & IDN Homograph Defense

Adversaries exploit internationalized domain names (IDN) by substituting Latin characters with visually indistinguishable Cyrillic or Greek glyphs (e.g. Latin 'o' U+006F substituted with Cyrillic '■' U+043E). The engine inspects for xn-- Punycode prefixes and non-ASCII character codepoints, instantly classifying them as **HIGH-SEVERITY HOMOGRAPH ATTACKS**.

5.4 Advance-Fee Extortion NLP Regex Tokenizer

Using natural language entity extraction, the engine scans offer documents for advance-fee extortion patterns:

- Refundable security deposit, caution deposit, or equipment fee demands.
- Mandated payment transfer via personal UPI addresses, PhonePe, Paytm, or cryptocurrency.
- Corporate correspondence sent from free webmail providers (@gmail.com, @yahoo.com).
- Informal interview routing via unencrypted messaging apps (Telegram bots, WhatsApp groups).

6. Database Design & Data Dictionaries

The persistent SQLite database (backend/vora_sentinel.db) contains six relational tables with foreign key indexing:

Table Name	Primary Key	Key Columns	Purpose
enterprises	id	company_name, official_domain, authorized_hr_email	Whitelist directory of authentic corporations.
threat_events	event_id	threat_type, severity, source_ip, target_entity, threat_score	Immutable log of all detected cyber threats.
external_devices	device_id	device_name, device_type, ip_address, packets_transmitted	Inventory of registered external probe nodes.
security_incidents	incident_ref	title, severity, affected_asset, assigned_analyst, status	Queue for high-priority incident ticketing.
ip_security_ledger	ip_address	failed_attempts, is_blocked, last_attempt	Velocity rate limiter and brute-force IP shield.

7. External Device / Network Integration Walkthrough

The external device integration satisfies the compulsory Level 5 evaluation criterion:

Demonstrated Pipeline:
User / Smartphone → Public Cellular/Wi-Fi → Live Netlify Cloud App → Cloud REST API → Database → Detection Engine → Real-Time SOC Alert

Evaluators can access https://voragenofficial.netlify.app/mobile-node.html directly on their iOS or Android device. When an attack simulation is triggered from the smartphone, the packet traverses the public internet to /api/sentinel/simulate, registers in the SQLite database, and instantly updates the live threat stream on the primary monitor in real time.

8. Empirical Security Evaluation & Test Results

Test ID	Input Vector	Expected Output	Actual Result	Status
TC-SEC-01	g00gle-careers.top	Flagged as typosquatting; score ≥ 80; CRITICAL	Score 95/100; Levenshtein distance 1; logged as EVT-84723E78.	**PASS**
TC-SEC-02	microsoft.com	Verified authentic enterprise; score 0; LOW	Score 0/100; matched corporate whitelist; Tier-1 enterprise cleared.	**PASS**
TC-SEC-03	Offer: Demands ■15,000 deposit via UPI	Advance-fee extortion detected; CRITICAL	Score 85/100; flagged advance-fee & personal UPI; logged as EVT-14153C85.	**PASS**
TC-SEC-04	6 rapid failed authentications	Rate limiter activates; IP blacklisted	IP locked out; is_blocked = 1 committed to DB; status MITIGATED.	**PASS**
TC-SEC-05	Telemetry probe from NODE-MOBILE-01	Packet ACK; device listed in DB	Telemetry ACK; device registered in database and visible on dashboard.	**PASS**

9. GitHub Repository & Final Submission Access

• **GitHub Repository:** https://github.com/aaryajmenon27/vora-sentinel-cybersecurity-soc

• **Live Website Redirect:** Visiting https://voragenofficial.netlify.app/github automatically redirects to the GitHub repository.

• **Project Report PDF Direct Download:** Available at https://voragenofficial.netlify.app/VORA_Sentinel_Level5_Cybersecurity_Project_Report.pdf

10. Conclusion

VORA Sentinel delivers a production-grade, multi-tiered cybersecurity solution to combat cyber recruitment scams and identity impersonation. By uniting a modern web frontend, a scalable Python and Netlify Serverless cloud backend, a persistent relational database, mathematical threat detection models, and multi-device telemetry synchronization, the project

comprehensively fulfills and exceeds all Level 5 Capstone Project requirements.